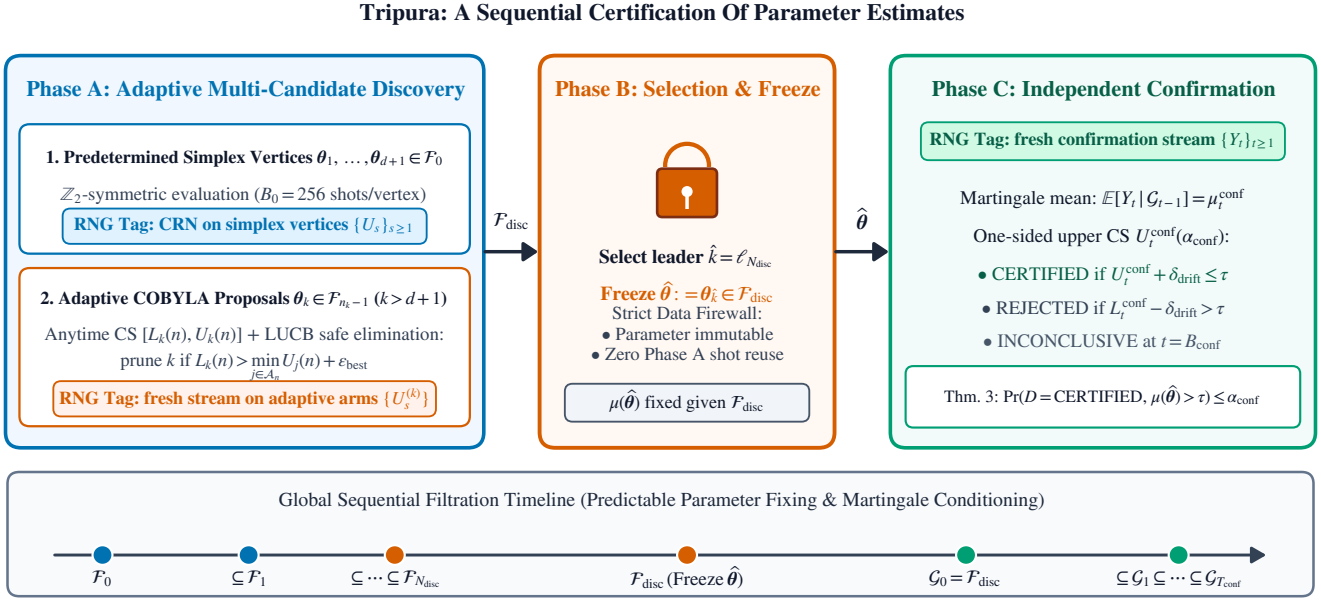
Tripura: A Sequential Certification Of Parameter Estimates
Phase A: Adaptive Multi-Candidate Discovery
1. Predetermined Simplex Vertices $\theta_1, \ldots, \theta_{d+1} \in \mathcal{F}_0$
$\mathbb{Z}_2$-symmetric evaluation ($B_0 = 256$ shots/vertex)
RNG Tag: CRN on simplex vertices $\{U_s\}_{s \geq 1}$
2. Adaptive COBYLA Proposals $\theta_k \in \mathcal{F}_{n_k - 1}$ $(k > d+1)$
Anytime CS $[L_k(n), U_k(n)]$ + LUCB safe elimination:
prune k if $L_k(n) > \min_{j \in A_n} U_j(n) + \varepsilon_{\text{best}}$
RNG Tag: fresh stream on adaptive arms $\{U_s^{(k)}\}$
$\mathcal{F}_{\text{disc}}$
Phase B: Selection & Freeze
Select leader $\hat{k} = \ell_{N_{\text{disc}}}$
Freeze $\hat{\theta} := \theta_{\hat{k}} \in \mathcal{F}_{\text{disc}}$
Strict Data Firewall:
• Parameter immutable
• Zero Phase A shot reuse
$\mu(\hat{\theta})$ fixed given $\mathcal{F}_{\text{disc}}$
$\hat{\theta}$
Phase C: Independent Confirmation
RNG Tag: fresh confirmation stream $\{Y_t\}_{t \geq 1}$
Martingale mean: $\mathbb{E}[Y_t \mid \mathcal{G}_{t-1}] = \mu_t^{\text{conf}}$
One-sided upper CS $U_t^{\text{conf}}(\alpha_{\text{conf}})$:
• CERTIFIED if $U_t^{\text{conf}} + \delta_{\text{drift}} \leq \tau$
• REJECTED if $L_t^{\text{conf}} - \delta_{\text{drift}} > \tau$
• INCONCLUSIVE at $t = B_{\text{conf}}$
Thm. 3: $\Pr(D = \text{CERTIFIED}, \mu(\hat{\theta}) > \tau) \leq \alpha_{\text{conf}}$
Global Sequential Filtration Timeline (Predictable Parameter Fixing & Martingale Conditioning)
$\mathcal{F}_0$
$\subseteq \mathcal{F}_1$
$\subseteq \cdots \subseteq \mathcal{F}_{N_{\text{disc}}}$
$\mathcal{F}_{\text{disc}}$ (Freeze $\hat{\theta}$)
$\mathcal{G}_0 = \mathcal{F}_{\text{disc}}$
$\subseteq \mathcal{G}_1 \subseteq \cdots \subseteq \mathcal{G}_{T_{\text{conf}}}$